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(54) **APPARATUS AND METHOD FOR MEDIA STREAMING**

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H04L 65/75 (2022.01)
H04L 65/80 (2022.01)
H04N 19/115 (2014.01)
H04N 19/14 (2014.01)

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(58) **Field of Classification Search**

None

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,374,288 B1 4/2002 Bhagavath
6,801,947 B1 10/2004 Li

(Continued)

Primary Examiner — Joshua D Taylor

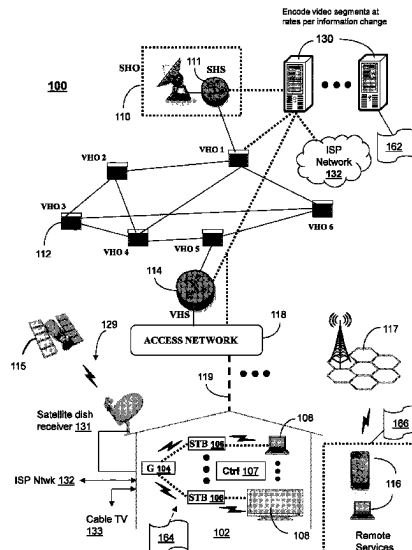
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ABSTRACT

A system that incorporates teachings of the subject disclosure may include, for example, dividing a media content item into a plurality of media content segments, at each media content segment of the plurality of media content segments, applying motion estimation between at least two video frames of the media content segment to determine a content weight for the media content segment, determining a bit rate for the media content segment according to the determined content weight for the media content segment, and encoding the media content segment at the determined bit rates to generate a data stream for the media content segment, whereby a plurality of data streams for the plurality of media content segments of the media content are generated, and transmitting the plurality of data streams for the plurality of media content segments of the media content to the one or more media devices. Other embodiments are disclosed.

20 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/513,924, filed on Jul. 17, 2019, now Pat. No. 11,184,681, which is a division of application No. 14/693,087, filed on Apr. 22, 2015, now Pat. No. 10,405,055, which is a continuation of application No. 13/455,801, filed on Apr. 25, 2012, now Pat. No. 9,042,441.

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H04N 21/2381 (2011.01)

H04N 21/2383 (2011.01)

H04N 21/643 (2011.01)

(52) **U.S. Cl.**

CPC ... **H04N 21/2383** (2013.01); **H04N 21/64322** (2013.01)

(56)

References Cited

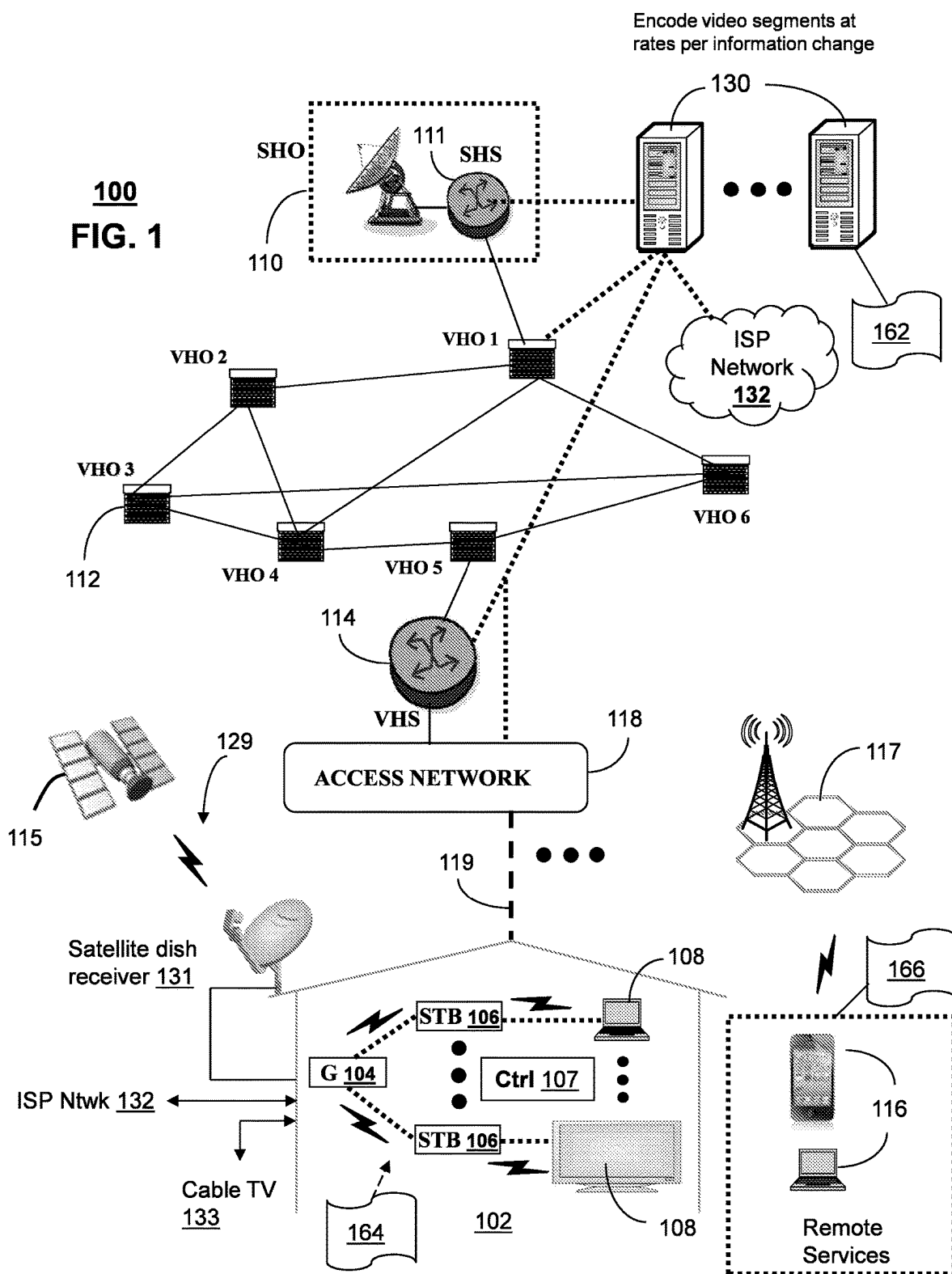
U.S. PATENT DOCUMENTS

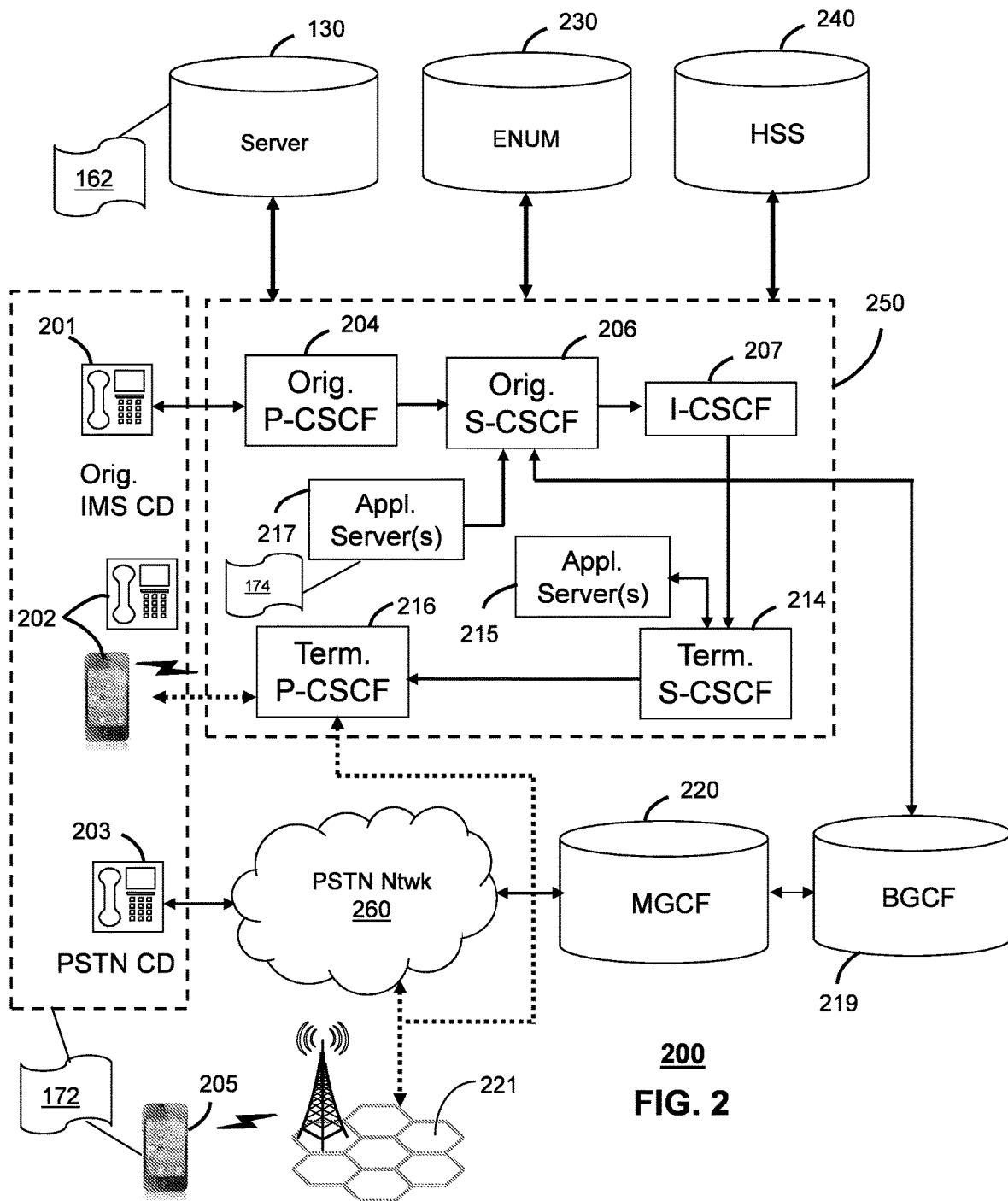
9,615,098 B1 * 4/2017 Yi H04N 19/14
 11,184,681 B2 11/2021 Shaw et al.
 11,659,253 B2 * 5/2023 Shaw H04L 65/762
 725/109
 2002/0129378 A1 * 9/2002 Cloonan H04N 21/266
 348/E7.071

2002/0131496 A1 9/2002 Vasudevan et al.
 2004/0120411 A1 6/2004 Walton et al.
 2005/0122999 A1 6/2005 Scherzer et al.
 2006/0171457 A1 8/2006 DeGarrido
 2006/0204113 A1 9/2006 Wang
 2006/0222078 A1 10/2006 Raveendran
 2007/0074266 A1 * 3/2007 Raveendran H04N 19/172
 725/94
 2007/0204067 A1 8/2007 Walker et al.
 2008/0025211 A1 1/2008 Karaoguz et al.
 2008/0059646 A1 * 3/2008 Ludvig H04L 69/14
 709/231
 2008/0092181 A1 * 4/2008 Britt H04N 21/482
 348/E7.071
 2008/0165861 A1 7/2008 Wen et al.
 2008/0201748 A1 8/2008 Hasek et al.
 2010/0014528 A1 1/2010 Amir et al.
 2010/0189183 A1 7/2010 Gu et al.
 2010/0272173 A1 10/2010 Puri et al.
 2011/0030019 A1 2/2011 Ulm
 2011/0179186 A1 7/2011 Li et al.
 2012/0141089 A1 * 6/2012 Hunt H04N 21/2385
 386/239
 2012/0195369 A1 8/2012 Guerrero
 2019/0342627 A1 11/2019 Shaw et al.
 2022/0046330 A1 2/2022 Shaw et al.

* cited by examiner

100
FIG. 1





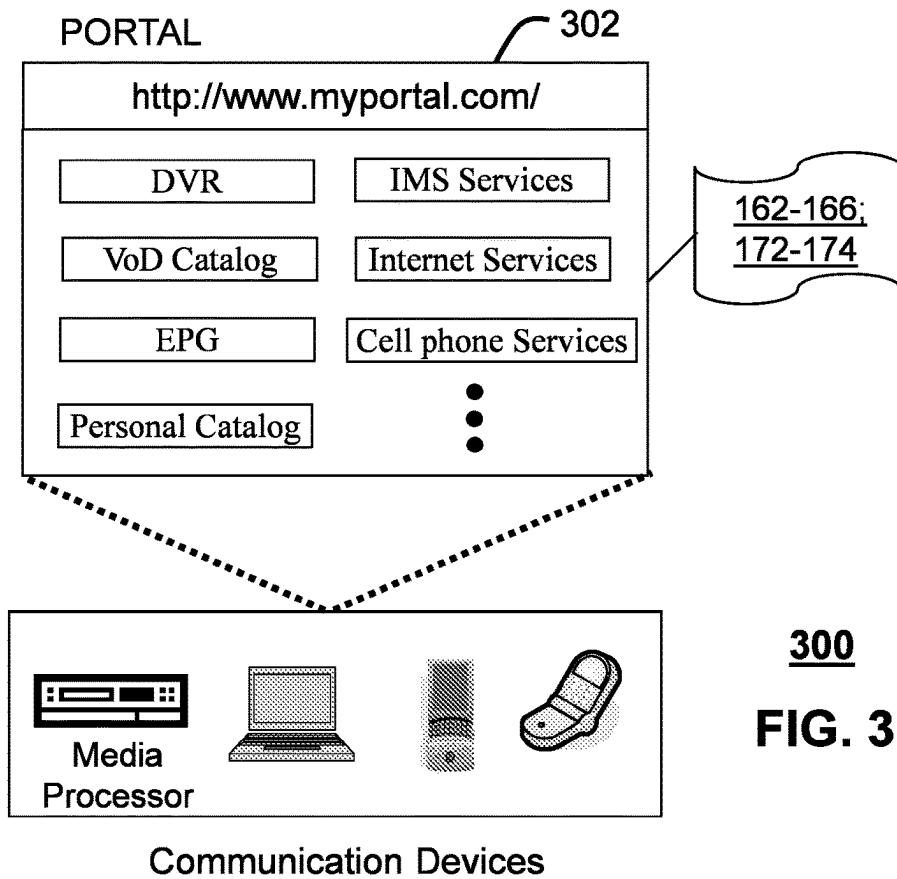


FIG. 3

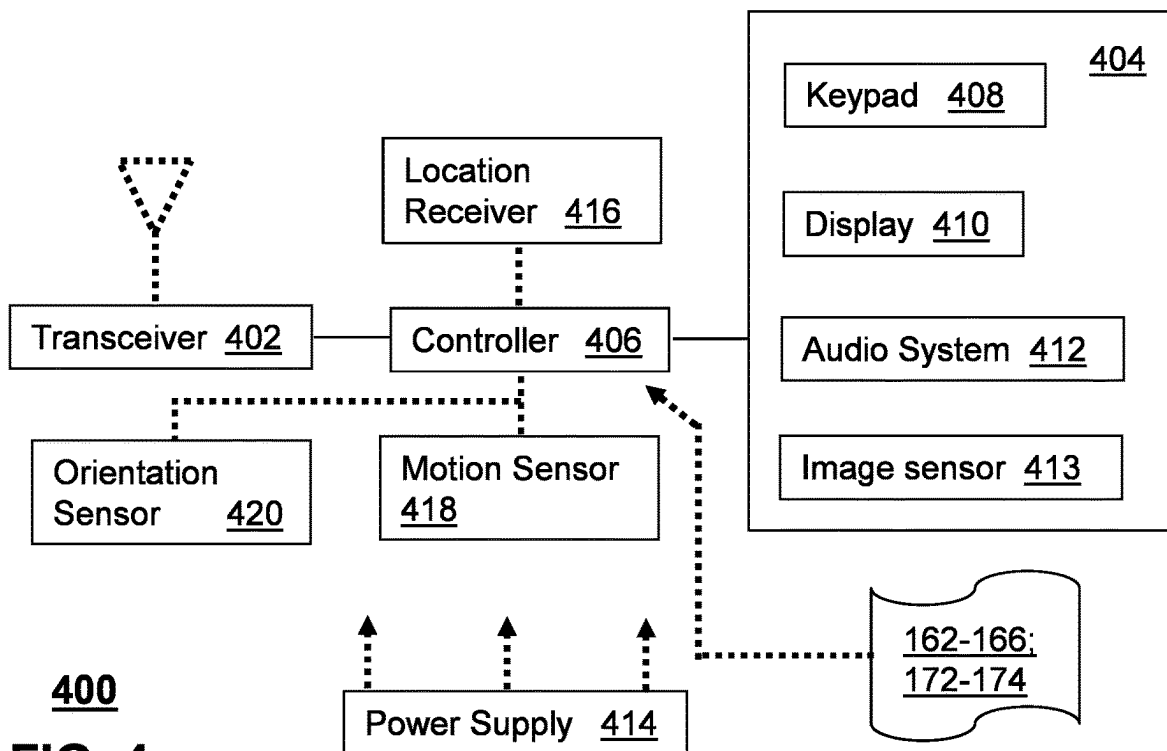
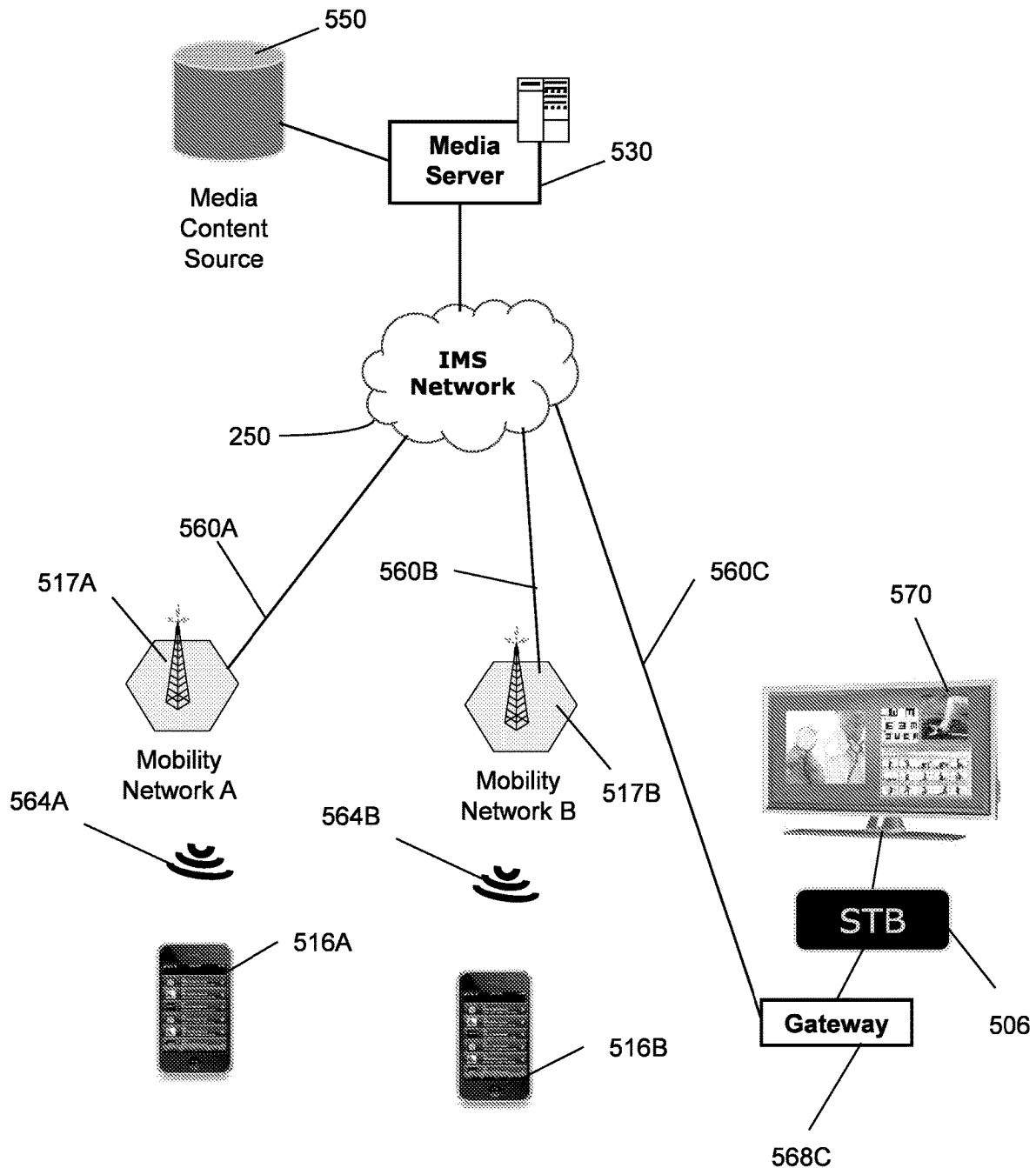
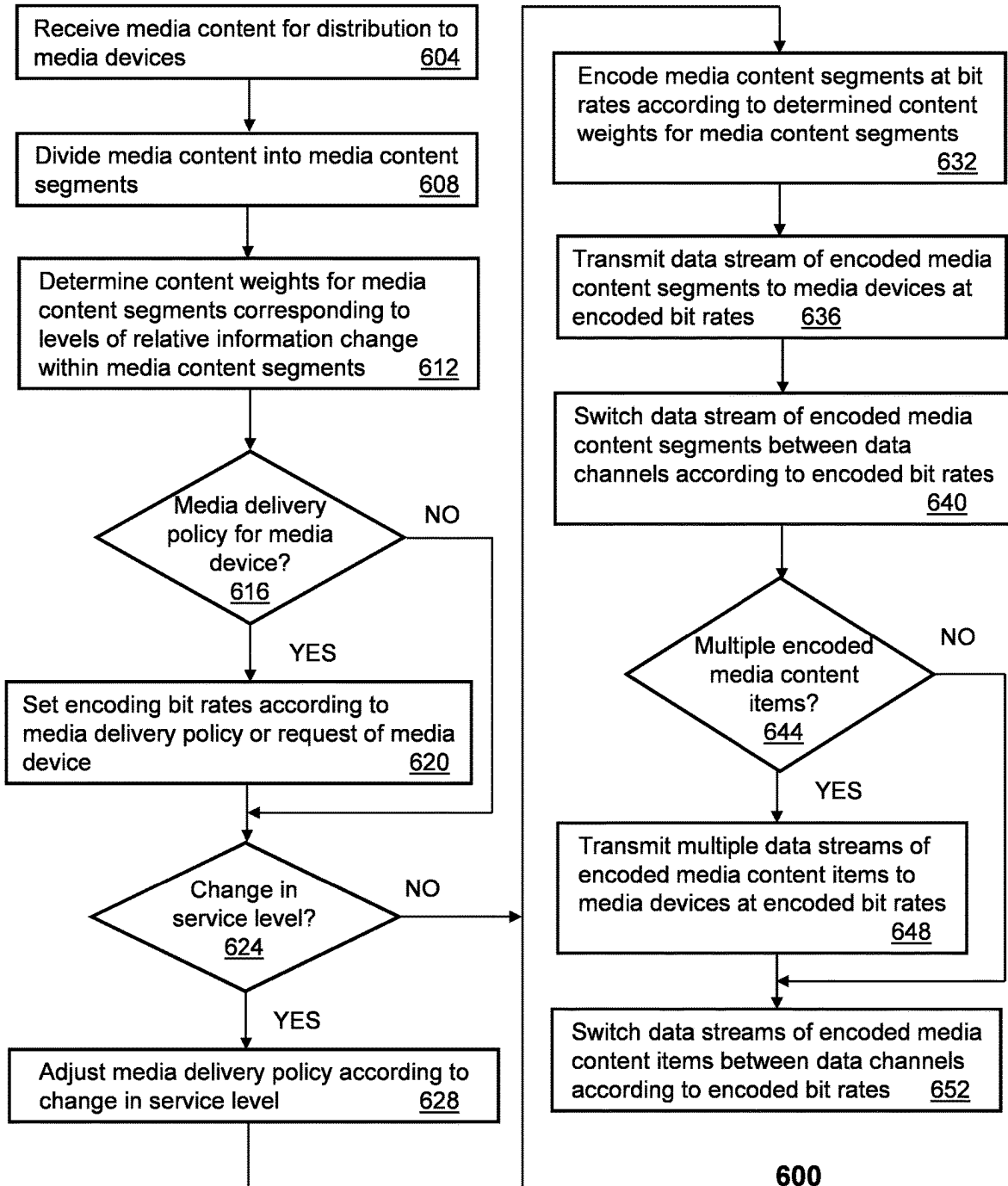


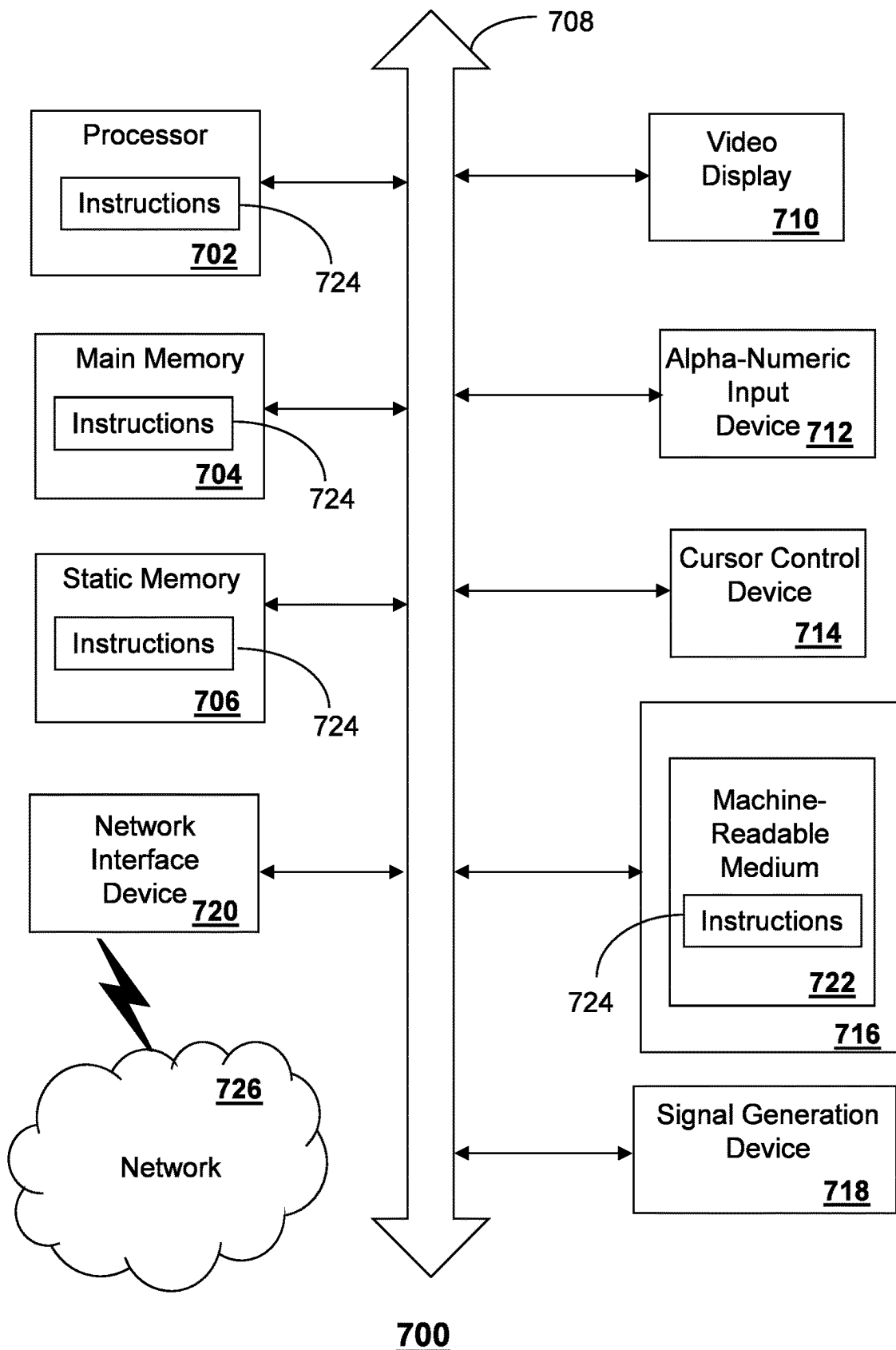
FIG. 4



500
FIG. 5



600
FIG. 6

**FIG. 7**

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APPARATUS AND METHOD FOR MEDIA STREAMING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/511,221, filed Oct. 26, 2021, which is a continuation of U.S. application Ser. No. 16/513,924, filed Jul. 17, 2019 (now U.S. Pat. No. 11,184,681), which is a divisional of U.S. application Ser. No. 14/693,087, filed Apr. 22, 2015 (now U.S. Pat. No. 10,405,055), which is a continuation of U.S. application Ser. No. 13/455,801, filed Apr. 25, 2012 (now U.S. Pat. No. 9,042,441), which are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

The subject disclosure relates generally to media content and more specifically to an apparatus and method for video streaming.

BACKGROUND

Media content is frequently experienced by consumers via devices such as computers, televisions, radios, and mobile electronics. Media content is frequently delivered by service providers, who send the media content, such as television, radio, and video programming, directly to consumers for enjoyment at their physical locations. Modern communication networks benefit from interconnectivity between consumers and various communication devices. As network capabilities expand, these interconnections provide new opportunities to enhance abilities to enjoy media content.

BRIEF DESCRIPTION OF THE DRAWINGS

Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

FIGS. 1-2 depict illustrative embodiments of communication systems that provide media services;

FIG. 3 depicts an illustrative embodiment of a web portal for interacting with the communication systems of FIGS. 1-2;

FIG. 4 depicts an illustrative embodiment of a communication device utilized in the communication systems of FIGS. 1-2;

FIG. 5 depicts an illustrative embodiment of a communication system that performs streaming of media content;

FIG. 6 depicts an illustrative embodiment of a method operating in portions of the systems described in FIGS. 1-5; and

FIG. 7 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions, when executed, may cause the machine to perform any one or more of the methods described herein.

DETAILED DESCRIPTION

The subject disclosure describes, among other things, illustrative embodiments of a system and a method of streaming media content, where the streamed media content is encoded at bit rates according to the content weight of the media content. Other embodiments are contemplated by the subject disclosure.

One embodiment of the subject disclosure includes a memory storing computer instructions and a processor

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coupled to the memory. The processor can perform operations responsive to executing the computer instructions including receiving media content for distribution to one or more media devices. The processor can perform operations for dividing the received media content into a plurality of media content segments. For each media content segment of the plurality of media content segments, the processor can perform operations for determining a content weight for the media content segment from motion estimation between at least two video frames of the media content segment, for determining a bit rate for the media content segment according to the determined content weight for the media content segment, and for encoding the media content segment at the determined bit rate for the media content segment to generate a data stream for the media content segment. A plurality of data streams for the plurality of media content segments of the media content can thereby be generated. The processor can, in turn, perform operations for transmitting the plurality of data streams for the plurality of media content segments of the media content to the one or more media devices.

One embodiment of the subject disclosure includes computer instructions, which, responsive to being executed by at least one processor, can cause the at least one processor to perform operations including dividing media content into a plurality of media content segments. For each media content segment of the plurality of media content segments, the computer instructions can cause the at least one processor to perform operations for determining a content weight for the media content segment, for determining a bit rate for the media content segment according to the determined content weight for the media content segment, and for encoding the media content segment at the determined bit rate for the media content segment to generate a data stream for the media content segment. A plurality of data stream for the plurality of media content segments can thereby be generated. The computer instructions can cause the at least one processor to perform operations for transmitting the plurality of data streams for the plurality of media content segments of the media content to one or more media devices over a plurality of data channels. The computer instructions can, in turn, cause the at least one processor to perform operations for switching the plurality of data streams between the plurality of data channels according to at least one determined bit rate for at least one media content segment of the plurality of media content segments.

One embodiment of the subject disclosure includes a method including utilizing a system including at least one processor for dividing a media content item into a plurality of media content segments. At each media content segment of the plurality of media content segments, the system can be utilized for applying motion estimation between at least two video frames of the media content segment to determine a content weight for the media content segment, determining a bit rate for the media content segment according to the determined content weight for the media content segment, and encoding the media content segment at the determined bit rates to generate a data stream for the media content segment. A plurality of data streams for the plurality of media content segments of the media content can thereby be generated. The system can, in turn, be utilized for transmitting the plurality of data streams for the plurality of media content segments of the media content to the one or more media devices.

FIG. 1 depicts an illustrative embodiment of a first communication system 100 for delivering media content. The communication system 100 can represent an Internet

Protocol Television (IPTV) media system. Communication system **100** can also provide for all or a portion of the computing devices **130** to function as a media server (herein referred to as media server **130**). The media server **130** can use computing and communication technology to perform function **162**, which can include among things, receive media content from, for example, the super headend office server **111** for distribution through the system **100**. The media server **130** can divide the received media content into one or more media content segments. The media server **130** can then determine content weights for each of the media content segments, where the content weights correspond to levels of relative information change within each of the media content segments. The media server **130** can encode the media content segments, where each of the media content segments is encoded at a bit rate according to the content weight for that media content segment. The media server **130** can then transmit the encoded media content segments one or more data streams to media devices by way of an access network **118**, where the media devices can include media processors **106** and wireless communication devices **116**. The media server **130** can further switch the transmission of data stream to different data channels within the access network **118** according to the bit rates of the encoded media content segments. The media processors **106** and wireless communication devices **116** can be adapted with software functions **164** and **166**, respectively, to utilize the services of media server **130**.

The IPTV media system can include a super head-end office (SHO) **110** with at least one super headend office server (SHS) **111** which receives media content from satellite and/or terrestrial communication systems. In the present context, media content can represent, for example, audio content, moving image content such as 2D or 3D videos, video games, virtual reality content, still image content, and combinations thereof. The SHS server **111** can forward packets associated with the media content to one or more video head-end servers (VHS) **114** via a network of video head-end offices (VHO) **112** according to a common multicast communication protocol.

The VHS **114** can distribute multimedia broadcast content via an access network **118** to commercial and/or residential buildings **102** housing a gateway **104** (such as a residential or commercial gateway). The access network **118** can represent a group of digital subscriber line access multiplexers (DSLAMs) located in a central office or a service area interface that provide broadband services over fiber optical links or copper twisted pairs **119** to buildings **102**. The gateway **104** can use common communication technology to distribute broadcast signals to media processors **106** such as Set-Top Boxes (STBs) which in turn present broadcast channels to media devices **108** such as computers or television sets managed in some instances by a media controller **107** (such as an infrared or RF remote controller).

The gateway **104**, the media processors **106**, and media devices **108** can utilize tethered communication technologies (such as coaxial, powerline or phone line wiring) or can operate over a wireless access protocol such as Wireless Fidelity (WiFi), Bluetooth, Zigbee, or other present or next generation local or personal area wireless network technologies. By way of these interfaces, unicast communications can also be invoked between the media processors **106** and subsystems of the IPTV media system for services such as video-on-demand (VoD), browsing an electronic programming guide (EPG), or other infrastructure services.

A satellite broadcast television system **129** can be used also in the media system of FIG. 1. The satellite broadcast

television system can be overlaid, operably coupled with, or replace the IPTV system as another representative embodiment of communication system **100**. In this embodiment, signals transmitted by a satellite **115** carrying media content can be received by a satellite dish receiver **131** coupled to the building **102**. Modulated signals received by the satellite dish receiver **131** can be transferred to the media processors **106** for demodulating, decoding, encoding, and/or distributing broadcast channels to the media devices **108**. The media processors **106** can be equipped with a broadband port to the ISP network **132** to enable interactive services such as VoD and EPG as described above.

In yet another embodiment, an analog or digital cable broadcast distribution system such as cable TV system **133** can be overlaid, operably coupled with, or replace the IPTV system and/or the satellite TV system as another representative embodiment of communication system **100**. In this embodiment, the cable TV system **133** can also provide Internet, telephony, and interactive media services.

It is contemplated that the subject disclosure can apply to other present or next generation over-the-air and/or landline media content services system. Some of the network elements of the IPTV media system can be coupled to one or more computing devices **130**, a portion of which can operate as a web server for providing web portal services over an Internet Service Provider (ISP) network **132** to wireline media devices **108** or wireless communication devices **116**.

It is further contemplated that multiple forms of media services can be offered to media devices over landline technologies such as those described above. Additionally, media services can be offered to media devices by way of a wireless access base station **117** operating according to common wireless access protocols such as Global System for Mobile or GSM, Code Division Multiple Access or CDMA, Time Division Multiple Access or TDMA, Universal Mobile Telecommunications or UMTS, World interoperability for Microwave or WiMAX, Software Defined Radio or SDR, Long Term Evolution or LTE, and so on. Other present and next generation wide area wireless network technologies are contemplated by the subject disclosure.

FIG. 2 depicts an illustrative embodiment of a communication system **200** employing an IP Multimedia Subsystem (IMS) network architecture to facilitate the combined services of circuit-switched and packet-switched systems. Communication system **200** can be overlaid or operably coupled with communication system **100** as another representative embodiment of communication system **100**.

Communication system **200** can comprise a Home Subscriber Server (HSS) **240**, a telephone Number Mapping (ENUM) server **230**, and other common network elements of an IMS network **250**. The IMS network **250** can establish communications between IMS-compliant communication devices (CDs) **201**, **202**, Public Switched Telephone Network (PSTN) CDs **203**, **205**, and combinations thereof by way of a Media Gateway Control Function (MGCF) **220** coupled to a PSTN network **260**. The MGCF **220** need not be used when a communication session involves IMS CD to IMS CD communications. A communication session involving at least one PSTN CD may utilize the MGCF **220**.

IMS CDs **201**, **202** can register with the IMS network **250** by contacting a Proxy Call Session Control Function (P-CSCF) which communicates with an interrogating CSCF (I-CSCF), which in turn, communicates with a Serving CSCF (S-CSCF) to register the CDs with the HSS **240**. To initiate a communication session between CDs, an originating IMS CD **201** can submit a Session Initiation Protocol

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(SIP INVITE) message to an originating P-CSCF **204** which communicates with a corresponding originating S-CSCF **206**. The originating S-CSCF **206** can submit the SIP INVITE message to one or more application servers (ASs) **217** that can provide a variety of services to IMS subscribers.

For example, the application servers **217** can be used to perform originating call feature treatment functions on the calling party number received by the originating S-CSCF **206** in the SIP INVITE message. Originating treatment functions can include determining whether the calling party number has international calling services, call ID blocking, calling name blocking, 7-digit dialing, and/or is requesting special telephony features (e.g., *72 forward calls, *73 cancel call forwarding, *67 for caller ID blocking, and so on). Based on initial filter criteria (iFCs) in a subscriber profile associated with a CD, one or more application servers may be invoked to provide various call originating feature services.

Additionally, the originating S-CSCF **206** can submit queries to the ENUM system **230** to translate an E.164 telephone number in the SIP INVITE message to a SIP Uniform Resource Identifier (URI) if the terminating communication device is IMS-compliant. The SIP URI can be used by an Interrogating CSCF (I-CSCF) **207** to submit a query to the HSS **240** to identify a terminating S-CSCF **214** associated with a terminating IMS CD such as reference **202**. Once identified, the I-CSCF **207** can submit the SIP INVITE message to the terminating S-CSCF **214**. The terminating S-CSCF **214** can then identify a terminating P-CSCF **216** associated with the terminating CD **202**. The P-CSCF **216** may then signal the CD **202** to establish Voice over Internet Protocol (VoIP) communication services, thereby enabling the calling and called parties to engage in voice and/or data communications. Based on the iFCs in the subscriber profile, one or more application servers may be invoked to provide various call terminating feature services, such as call forwarding, do not disturb, music tones, simultaneous ringing, sequential ringing, etc.

In some instances the aforementioned communication process is symmetrical. Accordingly, the terms "originating" and "terminating" in FIG. 2 may be interchangeable. It is further noted that communication system **200** can be adapted to support video conferencing. In addition, communication system **200** can be adapted to provide the IMS CDs **201**, **202** with the multimedia and Internet services of communication system **100** of FIG. 1.

If the terminating communication device is instead a PSTN CD such as CD **203** or CD **205** (in instances where the cellular phone only supports circuit-switched voice communications), the ENUM system **230** can respond with an unsuccessful address resolution which can cause the originating S-CSCF **206** to forward the call to the MGCF **220** via a Breakout Gateway Control Function (BGCF) **219**. The MGCF **220** can then initiate the call to the terminating PSTN CD over the PSTN network **260** to enable the calling and called parties to engage in voice and/or data communications.

It is further appreciated that the CDs of FIG. 2 can operate as wireline or wireless devices. For example, the CDs of FIG. 2 can be communicatively coupled to a cellular base station **221**, a femtocell, a WiFi router, a DECT base unit, or another suitable wireless access unit to establish communications with the IMS network **250** of FIG. 2. The cellular access base station **221** can operate according to common wireless access protocols such as Global System for Mobile (GSM), Code Division Multiple Access (CDMA), Time

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Division Multiple Access (TDMA), Universal Mobile Telecommunications (UMTS), World interoperability for Microwave (WiMAX), Software Defined Radio (SDR), Long Term Evolution (LTE), and so on. Other present and next generation wireless network technologies are contemplated by the subject disclosure. Accordingly, multiple wireline and wireless communication technologies are contemplated for the CDs of FIG. 2.

It is further contemplated that cellular phones supporting LTE can support packet-switched voice and packet-switched data communications and thus may operate as IMS-compliant mobile devices. In this embodiment, the cellular base station **221** may communicate directly with the IMS network **250** as shown by the arrow connecting the cellular base station **221** and the P-CSCF **216**.

It is further understood that alternative forms of a CSCF can operate in a device, system, component, or other form of centralized or distributed hardware and/or software. Indeed, a respective CSCF may be embodied as a respective CSCF system having one or more computers or servers, either centralized or distributed, where each computer or server may be configured to perform or provide, in whole or in part, any method, step, or functionality described herein in accordance with a respective CSCF. Likewise, other functions, servers and computers described herein, including but not limited to, the HSS and ENUM server, the BGCF, and the MGCF, can be embodied in a respective system having one or more computers or servers, either centralized or distributed, where each computer or server may be configured to perform or provide, in whole or in part, any method, step, or functionality described herein in accordance with a respective function, server, or computer.

The media server **130** of FIG. 1 can be operably coupled to the second communication system **200** for purposes similar to those described above. It is further contemplated by the subject disclosure that media server **130** can perform function **162** and thereby provide streaming media content services to the CDs **201**, **202**, **203** and **205** of FIG. 2. CDs **201**, **202**, **203** and **205**, which can be adapted with software to perform function **172** to utilize the services of the media server **130**. It is further contemplated that the media server **130** can be an integral part of the application server(s) **217** performing function **174**, which can be substantially similar to function **162** and adapted to the operations of the IMS network **250**.

FIG. 3 depicts an illustrative embodiment of a web portal **302** which can be hosted by server applications operating from the computing devices **130** of the communication system **100** illustrated in FIG. 1. The web portal **302** can be used for managing services of communication systems **100-200**. A web page of the web portal **302** can be accessed by a Uniform Resource Locator (URL) with an Internet browser such as Microsoft's Internet Explorer™, Mozilla's Firefox™, Apple's Safari™, or Google's Chrome™ using an Internet-capable communication device such as those described in FIGS. 1-2. The web portal **302** can be configured, for example, to access a media processor **106** and services managed thereby such as a Digital Video Recorder (DVR), a Video on Demand (VoD) catalog, an Electronic Programming Guide (EPG), or a personal catalog (such as personal videos, pictures, audio recordings, etc.) stored at the media processor **106**. The web portal **302** can also be used for provisioning IMS services described earlier, provisioning Internet services, provisioning cellular phone services, and so on.

It is contemplated by the subject disclosure that the web portal **302** can further be utilized to manage and provision

software applications **162-166**, and **172-174**, such as transmitting and/or receiving streamed media content, and to adapt these applications as may be desired by subscribers and service providers of communication systems **100-200**.

FIG. **4** depicts an illustrative embodiment of a communication device **400**. Communication device **400** can serve in whole or in part as an illustrative embodiment of the devices depicted in FIGS. **1-2**. The communication device **400** can comprise a wireline and/or wireless transceiver **402** (herein transceiver **402**), a user interface (UI) **404**, a power supply **414**, a location receiver **416**, a motion sensor **418**, an orientation sensor **420**, and a controller **406** for managing operations thereof. The transceiver **402** can support short-range or long-range wireless access technologies such as Bluetooth, ZigBee, WiFi, Digital Enhanced Cordless Telecommunications (DECT), or cellular communication technologies, just to mention a few. Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, software defined radio (SDR), Long Term Evolution (LTE), as well as other next generation wireless communication technologies as they arise. The transceiver **402** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

The UI **404** can include a depressible or touch-sensitive keypad **408** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **400**. The keypad **408** can be an integral part of a housing assembly of the communication device **400** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth. The keypad **408** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **404** can further include a display **410** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **400**. In an embodiment where the display **410** is touch-sensitive, a portion or all of the keypad **408** can be presented by way of the display **410** with navigation features.

The display **410** can use touch screen technology to also serve as a user interface for detecting user input (e.g., touch of a user's finger). As a touch screen display, the communication device **400** can be adapted to present a user interface with graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The touch screen display **410** can be equipped with capacitive, resistive or other forms of sensing technology to detect much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used control the manipulation of the GUI elements.

The UI **404** can also include an audio system **412** that utilizes common audio technology for conveying low volume audio (such as audio heard only in the proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **412** can further include a microphone for receiving audible signals of an end user. The audio system **412** can also be used for voice recognition applications. The UI **404** can further include an image sensor **413** such as a charged coupled device (CCD) camera for capturing still or moving images.

The power supply **414** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and charging sys-

tem technologies for supplying energy to the components of the communication device **400** to facilitate long-range or short-range portable applications. Alternatively, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port. The location receiver **416** can utilize common location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **400** based on signals generated by a constellation of GPS satellites, thereby facilitating common location services such as navigation. The motion sensor **418** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing to detect motion of the communication device **400** in three-dimensional space. The orientation sensor **420** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **400** (North, South, West, East, combined orientations thereof in degrees, minutes, or other suitable orientation metrics).

The communication device **400** can use the transceiver **402** to also determine a proximity to a cellular, WiFi, Bluetooth, or other wireless access points by common sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or a signal time of arrival (TOA) or time of flight (TOF). The controller **406** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies.

Other components not shown in FIG. **4** are contemplated by the subject disclosure. For instance, the communication device **400** can include a reset button (not shown). The reset button can be used to reset the controller **406** of the communication device **400**. In yet another embodiment, the communication device **400** can also include a factory default setting button positioned below a small hole in a housing assembly of the communication device **400** to force the communication device **400** to re-establish factory settings. In this embodiment, a user can use a protruding object such as a pen or paper clip tip to reach into the hole and depress the default setting button.

The communication device **400** as described herein can operate with more or less components described in FIG. **4**. These variant embodiments are contemplated by the subject disclosure.

The communication device **400** can be adapted to perform the functions of the media processor **106**, the media devices **108**, or the portable communication devices **116** of FIG. **1**, as well as the IMS CDs **201-202** and PSTN CDs **203-205** of FIG. **2**. It will be appreciated that the communication device **400** can also represent other common devices that can operate in communication systems **100-200** of FIGS. **1-2** such as a gaming console and a media player.

It is contemplated by the subject disclosure that the communication device **400** shown in FIG. **4** or portions thereof can serve as a representation of one or more of the devices of communication systems **100-200**. It is further contemplated that the controller **406** can be adapted in various embodiments to perform the functions **162-166** and **172-174**, such as transmitting and/or receiving streamed media content.

FIG. **5** depicts an illustrative embodiment of a communication system **500** for providing media content. System **500** can be overlaid or operably coupled to communication systems **100-200** as another representative embodiment of communication systems **100-200**. System **500** can include a media server **530** in communication with a packet switched

network such as IMS network **250**. The media server **530** can be in communication with a media processor **506**, such as set-top box, that is further in communication with a media device **570**, such as a television. The media server **530** can provide media content, by way of a media stream, to the media processor device **506**. The media processor device **506** can further be in communication with mobile communication device **516A** and **516B**. The mobile communication devices **516A** and **516B** can be communicatively coupled to the IMS network **250** through base station **517A** of a mobility network A and base station **517B** of mobility network B. The mobility communication device **516A** and **516B** can communicate to the mobility network through a cellular communication technology link **545**. The mobile communication devices **516A** and **516B** can receive streamed media content from the media server device **530** through the IMS network **250** and the mobility networks **517A** and **517B**.

The media server **530** can receive media content from a media content source **550**. The media server **530** can divide the received media content into one or media content segments. The media server **530** can determine content weights for each of the media content segments, where the content weights correspond to levels of relative information change within each of the media content segments. The media server **530** can encode the media content segments, where each of the media content segments is encoded at a bit rate according to the content weight for that media content segment. The media server **530** can transmit the encoded media content segments as one or more data streams to media devices by way of an IMS network **250**, where the media devices can include media processors **506** and mobile communication devices **516A** and **516B**.

The media server **530** can switch the transmission of data streams to different data channels according to the bit rates of the encoded media content segments. For example, mobile communication device **516A** can be coupled to the media server **530** by way of data link **560A**, mobility network A **517A**, and wireless link **564A**. Similarly, mobile communication device **516B** can be coupled to the media server **530** by way of data link **560B**, mobility network B **517B**, and wireless link **564B**. Media processor device **506** can be coupled to the media server **530** by way of data link **560C** and gateway **568C**. The data links **560A**, **560B**, and **560C**, each can be made up of one or more data channels capable of transmitting streamed media content sourced from the media server **530** to the media devices **516A**, **516B**, and **506**.

The media server **530** and/or the IMS network **250** and/or mobility networks **517A** and **517B** can switch data streams between data channels within the data links based on the encoded bit rate of the data stream. In one embodiment, individual data channels within the data links, such as within data link **560A**, have different bandwidth capabilities. In another embodiment, individual data channels will experience different bandwidth loading as the data channels are used to communication information for several different devices, such as where the IMS network **250** is communicating with several different devices through the data link **560A** and the mobility network A **517A**. As the media server **530** encodes requested media content at various bit rates at different segments of the media content in response to variations in the media content weight, this streamed media content from the media server will exhibit corresponding changes in bit rate. For example, during an action sequence of a media program, the media server **530** may detect a high content weight in a segment of the media program—indi-

cating a large amount of variation in video image—and, in response, encode the media content at a high bit rate or, alternatively, a low compression rate. The media server **530** can then stream this encoded media segment at a high bit rate to the receiving device, such as the mobile communications device **516A**, over the data link **560A** and mobility network A **517A**. Some of the data channels of the data link **560A** and/or the mobility network A **517A** can be capable of carrying the high bit rate stream of the encoded segment due to higher inherent bandwidth or due to lower temporal loading conditions, while other data channels of the data link **560A** and/or the mobility network A **517A** can be incapable of carrying the high bit rate segment. The media server **530** and/or IMS Network **250** and/or the mobility network A **517A** can switch the high bit rate data stream between data channels of to compensate for loading conditions and/or inherent data channel limitations. The media server **530** can thereby provide streaming media content to the mobile communication devices **516A** and **516B** and to the media processor device **506** according to the illustrative embodiments of method **600** of FIG. **6**.

FIG. **6** depicts an illustrative method **600** that operates in portions of the devices of FIGS. **1-5**. Method **600** can begin with step **604** in which the media server **530** can receive media content for distribution to media devices. In one embodiment, the media content can be received from a super headend office server. In one embodiment, the media content can be broadcast content that is received by way of an over-the-air source such as radio-frequency broadcasting or satellite broadcasting. In one embodiment, the media content is received from a wired source, such as a cable system. In one embodiment, the media content can be video-on-demand programming, such as movies, special events, or television episodes that are recorded and stored at the communication system **100** and then accessed by the media server **530**.

In step **608**, the media server **530** can divide the received media content in to media content segments. In one embodiment, the media server **530** can divide the media content into media content segments of standard time periods. In one embodiment, the media server **530** can divide the media content according to content type, where each segment includes content of a single type. In one embodiment, the media server **530** can forgo dividing the media content and analyze the media content in its entirety.

In step **612**, the media server **530** can determine content weights for each of the media content segments. In one embodiment, the content weights correspond to levels of relative information change within each media content segment. In one embodiment, a fixed number of content weights can be used to characterize the relative level of information change in a media content segment. For example, three levels of content weights may be defined as low, medium, and high. For example, a media segment featuring mainly a talking head and largely an unmoving background—such as in a newscast—can be graded by the media server **530** as a low content weight. In another example, a segment featuring a slow motion video image can similarly be determined to be a low content weight segment by the media server **530**. By contrast, a video segment of a true-speed sporting event can be determined to be a high weight content segment.

In one embodiment, the media server **530** can use a motion estimation process to determine the content weight of media segments. For example, the media server **530** can determine a series of motion vectors for adjacent frames in a sequence of frames in a media segment. The motion vectors can describe transformational movements of objects

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in adjacent video frames as, for example, one frame image moves with respect to a first location at a first frame and a second location at a second, adjacent frame. Motion estimation can be combined with motion compensation as part of a process to compress the video media content. However, the media server **530** can calculate motion vectors for adjacent video frames and use the magnitude, or a statistic based from the magnitude, to represent a measure of the relative level of information change occurring in the media content images in the media segment.

In step **616**, the media server **530** can detect if the media device **516A**, **516B**, or **506** has a media delivery policy. In one embodiment, the media device can have a media delivery policy that specifies a bit rate to be used for the delivery of streamed media content to the media device. For example, a mobile communication device **516B** can specify a maximum bit rate that the device **516B** can reliably receive and reproduce at its display. In another example, the mobile communication device **516B** can specify a preferred bit rate for streamed content. In one embodiment, the media server **530** establishes and maintains delivery policies.

In step **620**, if the media device **516A**, **516B**, or **506** has a media delivery policy, then the media server **530** can set bit rates for encoding the media content segments according to the media policy. In one embodiment, the media policy can specify that the media server **530** will always deliver media content to the media device **516A** according to the slowest typical bit rate or always according to the fastest typical bit rate. For example, a service plan for a media device **516A** could specify low cost streaming of media content. In such a plan, the media server **530** can send low content weighted video segments at the lowest possible bit rate while sending high content weighted video segments at only a medium bit rate to save system resources. In a different service plan, the media server **530** can stream media content to a media device **506** at the highest required bit rate for the content weight. In this case, if the media server **530** determines that a high content weight is needed for a media content segment, such as a segment from an action film, then the media server **530** will stream the content even if this taxes system resources.

In step **624**, the media server can detect a change in a service level for the media device **516A**. In one example, the media device **516A** can request a change in service by submitting a request through a portal **302**. In one embodiment, a higher level of service can be purchased by the user of a media device **516A**. Conversely, the user of the media device **516B** could choose to reduce the service level. In step **628**, if there is a change in the service level for the media device **516A**, **516B**, or **506**, then the media server **530** can adjust the media delivery policy for the media device **516A**, **516B**, or **506** according to the change in the service level.

In step **632**, the media server **530** can encode the media content segments at bit rates according to the determined content weights for the media content segments. In one embodiment, the media server **530** can encode the media content segments, where each of the media content segments is encoded at a bit rate according to the content weight for that media content segment. The media server **530** can transmit the encoded media content segments as one or more data streams to media devices by way of an IMS network **250**, where the media devices can include media processors **506** and mobile communication devices **516A** and **516B**. In one embodiment, the encoding can be performed by the media server **530** by applying different levels of digital compression to different media segments according to the content weights of those segments. In one embodiment, the

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encoding can be performed by the media server **530** by applying different display pixel resolutions to different segments of the at least a portion of the plurality of media content segments.

In step **636**, the media server **530** can transmit a data stream of encoded media content segments to the media device **516A**, **516B**, or **506** at the encoded bit rates. In step **640**, the media server **530** can switch the data stream of encoded media content segments between data channels according to the encoded bit rates. The media server **530** and/or the IMS network **250** and/or mobility networks **517A** and **517B** can switch data streams between data channels within the data links **560A-C** based on the encoded bit rate of the data stream. In one embodiment, individual data channels within the data links, such as within data link **560A**, have different bandwidth capability. In another embodiment, individual data channels will experience different bandwidth loading as the data channels are used to communication information for several different devices, such as where the IMS network **250** is communicating with several different devices through the data link **560A** and the mobility network A **517A**.

As the media server **530** encodes requested media content at various bit rates at different segments of the media content in response to variations in the media content weight, this streamed media content from the media server can exhibit corresponding changes in bit rate. For example, during an action sequence of a media program, the media server **530** may detect a high content weight in a segment of the media program—indicating a large amount of variation in video image—and, in response, encode the media content at a high bit rate or, alternatively, a low compression rate. The media server **530** can then stream this encoded media segment at a high bit rate to the receiving device, such as the mobile communications device **516A**, over the data link **560A** and mobility network A **517A**.

In step **644**, the media server **530** can determine if there are multiple encoded media content items. In one embodiment, the media server **530** can receive multiple items of content for distribution to the multiple media devices **516A**, **516B**, and **506**. For example, each of the media devices **516A**, **516B**, and **506** can request different media programs or movies from the communication system **100**. The media server **530** can then receive each of the requested media content programs from a media source **550**, determine content weights for each of the content segments of these programs, and encode each of these content segments according to the content weights.

In step **648**, if there are multiple encoded media content items, then the media server **530** can transmit the multiple data streams of the encoded media content items to the media devices **516A**, **516B**, or **506** at the encoded bit rates. In step **652**, the media server **530** can switch the multiple data streams of encoded media content segments between data channels according to the encoded bit rates. Some of the data channels of the data link **560A** and/or the mobility network A **517A** can be capable of carrying the high bit rate stream of the encoded segment due to higher inherent bandwidth or due to lower temporal loading conditions, while other data channels of the data link **560A** and/or the mobility network A **517A** can be incapable of carrying the high bit rate segment. The media server **530** and/or IMS Network **250** and/or the mobility network A **517A** can switch the high bit rate data stream between data channels of to compensate for loading conditions and/or inherent data channel limitations.

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Upon reviewing the aforementioned embodiments, it would be evident to an artisan with ordinary skill in the art that said embodiments can be modified, reduced, or enhanced without departing from the scope and spirit of the claims described below. For example, in one embodiment, a media device **516A** can store a delivery policy as a configuration and transmit the policy details to a media server **530** upon request. In another embodiment, a user of the media device **516A** can select or configure elements of a delivery policy into the device. In another embodiment, a media server **530** and/or another element of the communication system **100** can store delivery policy information for each device that has permission to receive media content from the media server **530**.

In one embodiment, a regular service plan would allow a media device **516B** to receive content streams at low and medium bit rates, while a premium service plan would allow streaming at low, medium, or high rates. In one embodiment, the media server **530** can set the encoding bit rates as requested by the media device **516A**, **516B**, or **506**. Other embodiments are contemplated by the subject disclosure.

FIG. 7 depicts an exemplary diagrammatic representation of a machine in the form of a computer system **700** within which a set of instructions, when executed, may cause the machine to perform any one or more of the methods discussed above. One or more instances of the machine can operate, for example, as the media server **130**, the media processor **106**, and/or the mobile communication device **516**, and/or other devices of FIGS. 1-5. In some embodiments, the machine may be connected (e.g., using a network) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client user machine in server-client user network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

The machine may comprise a server computer, a client user computer, a personal computer (PC), a tablet PC, a smart phone, a laptop computer, a desktop computer, a control system, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. It will be understood that a communication device of the subject disclosure includes broadly any electronic device that provides voice, video or data communication. Further, while a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods discussed herein.

The computer system **700** may include a processor **702** (e.g., a central processing unit (CPU), a graphics processing unit (GPU, or both), a main memory **704** and a static memory **706**, which communicate with each other via a bus **708**. The computer system **700** may further include a video display unit **710** (e.g., a liquid crystal display (LCD), a flat panel, or a solid state display. The computer system **700** may include an input device **712** (e.g., a keyboard), a cursor control device **714** (e.g., a mouse), a disk drive unit **716**, a signal generation device **718** (e.g., a speaker or remote control) and a network interface device **720**.

The disk drive unit **716** may include a tangible computer-readable storage medium **722** on which is stored one or more sets of instructions (e.g., software **724**) embodying any one or more of the methods or functions described herein, including those methods illustrated above. The instructions **724** may also reside, completely or at least partially, within the main memory **704**, the static memory **706**, and/or within

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the processor **702** during execution thereof by the computer system **700**. The main memory **704** and the processor **702** also may constitute tangible computer-readable storage media.

Dedicated hardware implementations including, but not limited to, application specific integrated circuits, programmable logic arrays and other hardware devices can likewise be constructed to implement the methods described herein. Applications that may include the apparatus and systems of various embodiments broadly include a variety of electronic and computer systems. Some embodiments implement functions in two or more specific interconnected hardware modules or devices with related control and data signals communicated between and through the modules, or as portions of an application-specific integrated circuit. Thus, the example system is applicable to software, firmware, and hardware implementations.

In accordance with various embodiments of the subject disclosure, the methods described herein are intended for operation as software programs running on a computer processor. Furthermore, software implementations can include, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing can also be constructed to implement the methods described herein.

While the tangible computer-readable storage medium **622** is shown in an example embodiment to be a single medium, the term "tangible computer-readable storage medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term "tangible computer-readable storage medium" shall also be taken to include any non-transitory medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methods of the subject disclosure.

The term "tangible computer-readable storage medium" shall accordingly be taken to include, but not be limited to: solid-state memories such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other re-writable (volatile) memories, a magneto-optical or optical medium such as a disk or tape, or other tangible media which can be used to store information. Accordingly, the disclosure is considered to include any one or more of a tangible computer-readable storage medium, as listed herein and including art-recognized equivalents and successor media, in which the software implementations herein are stored.

Although the present specification describes components and functions implemented in the embodiments with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. Each of the standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, and HTTP) represent examples of the state of the art. Such standards are from time-to-time superseded by faster or more efficient equivalents having essentially the same functions. Wireless standards for device detection (e.g., RFID), short-range communications (e.g., Bluetooth, WiFi, Zigbee), and long-range communications (e.g., WiMAX, GSM, CDMA, LTE) are contemplated for use by computer system **700**.

The illustrations of embodiments described herein are intended to provide a general understanding of the structure of various embodiments, and they are not intended to serve as a complete description of all the elements and features of apparatus and systems that might make use of the structures

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described herein. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. Figures are also merely representational and may not be drawn to scale. Certain proportions thereof may be exaggerated, while others may be minimized. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are contemplated by the subject disclosure.

The Abstract of the Disclosure is provided with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

What is claimed is:

1. A device comprising:

a processing system including a processor; and
a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, comprising:

determining a first content weight for a first media content segment using a motion estimation process for images in frames of the first media content segment, wherein the motion estimation process includes calculating a series of motion vectors for adjacent frames of the first media content segment, and wherein at least one of the series of motion vectors has a vector magnitude representing a relative level of information change occurring in the images;

encoding the first media content segment at a first bit rate according to a first content weight of the first media content segment by applying a first pixel resolution to the first media content segment to generate a first data stream;

transmitting, to a first media device via a first data channel, the first data stream associated with the first media content segment; and

switching transmission of the first data stream from the first data channel to a second data channel according to an available bandwidth at the second data channel.

2. The device of claim 1, wherein the operations further comprise transmitting a second data stream associated with a second media content segment encoded at a second bit rate according to a second content weight.

3. The device of claim 1, wherein the relative level of information change corresponds to a statistic based on the vector magnitude.

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4. The device of claim 1, wherein the first data stream is transmitted according to a first media delivery policy associated with the first media device.

5. The device of claim 4, wherein the first media delivery policy is determined according to a service level associated with the first media device.

6. The device of claim 5, wherein the operations further comprise:

detecting a change in the service level; and

updating the first media delivery policy responsive to the change in the service level.

7. The device of claim 1, wherein the operations further comprise determining the first bit rate in accordance with the first content weight.

8. The device of claim 1, wherein the operations further comprise receiving a request for a fixed bit rate for the first data stream, wherein the encoding of the first media content segment is performed according to the fixed bit rate.

9. A method comprising:

determining, by a processing system including a processor, a first content weight for a first media content segment using a motion estimation procedure for images in frames of the first media content segment, wherein the motion estimation procedure includes calculating a series of motion vectors for adjacent frames of the first media content segment, wherein at least one of the series of motion vectors has a vector magnitude representing a relative level of information change occurring in the images;

encoding, by the processing system, the first media content segment at a first bit rate according to a first content weight of the first media content segment by applying a first pixel resolution to the first media content segment to generate a first data stream;

transmitting, by the processing system to a first media device via a first data channel, the first data stream associated with the first media content segment; and

switching, by the processing system, transmission of the first data stream from the first data channel to a second data channel according to an available bandwidth at the second data channel.

10. The method of claim 9, further comprising transmitting a second data stream associated with a second media content segment encoded at a second bit rate according to a second content weight.

11. The method of claim 9, wherein the relative level of information change corresponds to a statistic based on the vector magnitude.

12. The method of claim 9, wherein the first data stream is transmitted according to a first media delivery policy associated with the first media device.

13. The method of claim 12, wherein the first media delivery policy is determined according to a service level associated with the first media device, and further comprising:

detecting, by the processing system, a change in the service level; and

updating, by the processing system, the first media delivery policy responsive to the change in the service level.

14. The method of claim 9, further comprising determining, by the processing system, the first bit rate in accordance with the first content weight.

15. The method of claim 9, further comprising receiving, by the processing system, a request for a fixed bit rate for the first data stream, wherein the encoding of the first media content segment is performed according to the fixed bit rate.

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16. A non-transitory machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, comprising:

determining a first content weight for a first media content segment by estimating motion of images in frames of the first media content segment, wherein the estimating includes calculating a series of motion vectors for adjacent frames of the first media content segment, wherein at least one of the series of motion vectors has a vector magnitude representing a relative level of information change occurring in the images;

encoding the first media content segment at a first bit rate according to a first content weight of the first media content segment by applying a first pixel resolution to the first media content segment to generate a first data stream;

transmitting, to a first media device via a first data channel, the first data stream associated with the first media content segment; and

switching transmission of the first data stream from the first data channel to a second data channel according to an available bandwidth at the second data channel.

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17. The non-transitory machine-readable medium of claim 16, wherein the operations further comprise transmitting a second data stream associated with a second media content encoded at a second bit rate according to a second content weight.

18. The non-transitory machine-readable medium of claim 17, wherein the relative level of information change corresponds to a statistic based on the vector magnitude.

19. The non-transitory machine-readable medium of claim 16, wherein the first data stream is transmitted according to a first media delivery policy associated with the first media device.

20. The non-transitory machine-readable medium of claim 19, wherein the first media delivery policy is determined according to a service level associated with the first media device, and wherein the operations further comprise: detecting a change in the service level; and updating the first media delivery policy responsive to the change in the service level.

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